

AYMANE AIT DADS

AI/ML Intern | PyTorch · Signal Processing · Benchmark Pipelines

Ayman.Ait-dads@eurecom.fr | +33 7 60 92 50 93 | linkedin.com/in/aymane-ait-dads | Sophia Antipolis, France

May 02, 2026

Hiring Team - AI/ML Internship

NextNav · Santa Clara, CA, USA

Dear NextNav Hiring Team,

Your role calls for someone who can construct **model training pipelines** against **high-precision ground-truth data** and deliver **reproducible benchmarking** results - exactly what I built at EURECOM when I ranked **#1 on the class leaderboard** by designing a rigorous ablation pipeline across transformer unfreezing depth, learning rate schedules, and augmentation strategies on 250K labeled samples, producing documented comparative results that proved 1-epoch fine-tuning generalizes better than extended re-training.

On the **deep learning training and optimization** side, I fine-tuned **CLIP ViT-L/14 (428M parameters)** using **FP16 mixed precision with GradScaler** under real GPU budget constraints, directly addressing the efficiency dimension your team requires. For signal-structured data, I built a **transformer autoencoder** operating on Mel spectrograms for unsupervised industrial fault detection, achieving **AUC 0.80** with no labeled anomalies - demonstrating the kind of **statistics and signal processing** background your requirements specify.

NextNav is solving a problem GPS alone cannot - delivering reliable 3D positioning in dense urban and indoor environments where signal degradation is the core challenge. That is a model accuracy and robustness problem, and it maps directly to what I do. I would welcome the chance to contribute from your Santa Clara office while collaborating with the Paris team - a cross-timezone dynamic I am well-positioned for as a French-based student fluent in both English and French.

Sincerely,

AYMANE AIT DADS